SPECIFICATION — NODE 1 — VIBRATION / HAPTIC ACTUATION — PAGE 1

DISCREET HAPTIC GUIDANCE, COUPLING CONFIRMATION, EVENT MARKING, QUALITY-GATED FEEDBACK, AND OPTIONAL AROUSAL ACTIVATION FOR AN ALL-DAY/NIGHT PHYSIOLOGY WEARABLE (NON-LIMITING)

A. Overview (Non-Limiting)

In various embodiments, Node 1 optionally includes a vibration or haptic actuator configured to provide non-visual, privacy-preserving feedback and/or user-enabled actuation modes during all-day/night physiologic monitoring. Node 1 is an intimate-region wearable device implemented as a C-shape ring or fully enclosed ring and configured to sense one or more physiologic parameters including temperature, pressure/contact, strain, hall effect and magnet events, optical pulse sensing (PPG), and motion/orientation via an inertial measurement unit (IMU). Haptic actuation is optional and may be used to improve setup correctness, wear stability, measurement repeatability, event tagging, quality awareness, and, in certain user-enabled modes, arousal activation or physiologic priming. Haptic actuation may be enabled or disabled and is not required for baseline sensing functionality.

B. Actuator Types (Non-Limiting)

Non-limiting actuator implementations include: 1. Eccentric Rotating Mass (ERM) motors; 2. Linear Resonant Actuators (LRA); 3. piezoelectric actuators (benders or stacks); 4. voice-coil actuators or micro-speaker haptics; and 5. any equivalent vibration-generating element capable of producing a perceivable pattern under low power. In various embodiments, the device includes a driver circuit configured for amplitude modulation, pattern timing, duty-cycle limiting, and safety bounds.

C. Mechanical Integration, Isolation, and Placement (Non-Limiting)

In various embodiments, the actuator is integrated within a thickened housing region, an electronics pod, or a distributed structure along the ring body. Non-limiting integration approaches include: 1. mounting the actuator in a localized pod positioned to transmit vibration through the ring body; 2. distributing vibration through structural elements to create a uniform haptic sensation; 3. incorporating damping and isolation (e.g., compliant mounts, elastomer layers, mass balancing) to reduce audible noise and reduce mechanical coupling into sensing regions; and 4. physically separating the actuator from sensitive sensing components to reduce vibration-induced measurement artifacts. In certain embodiments, the device uses soft overmold materials and internal damping to provide discreet haptics and reduce external audibility.

D. Haptic Purposes and Functional Modes (Non-Limiting)

In various embodiments, Node 1 uses haptics for one or more of the following purposes: 1. Fit / Coupling Confirmation: The device outputs a confirmation pattern when coupling plausibility is satisfied. Non-limiting coupling conditions include: pressure/contact thresholds met; stable strain baseline; adequate PPG coupling quality (e.g., signal amplitude/SNR); hall-effect/magnet closure confirmation for closure mechanisms; and IMU stability during a short validation window. 2. Guided Placement, Alignment, and Setup: The device outputs pulses that guide seating, closure, and alignment based on sensor plausibility checks and quality indices. Non-limiting guidance includes "adjust," "rotate," "close," or "hold still" cues provided as patterns. 3. Quality-Gated Feedback and Validity Awareness: The device provides discreet patterns indicating quality states such as valid/stable, low confidence, coupling degraded, excessive motion, or plausibility warnings. In certain embodiments, these cues are limited to setup or user-invoked modes to reduce disturbance during long wear. 4. Event Marking and Timeline Tagging: The device supports user-triggered event marking (e.g., via tap/gesture detected by IMU, magnet gesture, press sequence, or app command). The device provides haptic acknowledgment and inserts a time marker used to segment analysis windows (e.g., “start,” “stop,” “note,” “sleep attempt,” “session marker”) without requiring a screen interaction. 5. Discreet Alerts and Notifications: The device may provide silent notifications for battery state, pairing loss, memory/storage states, integrity faults, sensor faults, prolonged poor coupling, or user-configured alerts. 6. Optional Arousal Activation / Physiologic Priming Mode (User-Enabled, Non-Limiting): In various embodiments, Node 1 may optionally provide haptic actuation patterns intended for arousal activation, physiologic priming, or user-perceived stimulation cues. This mode may be user-initiated and consent-gated and may be time-limited and safety-bounded. In certain embodiments, actuation intervals are treated as actuation windows and are explicitly tagged to support correct interpretation of sensed responses.

E. Haptic Pattern Encoding (Non-Limiting)

In various embodiments, haptic output is encoded as patterns defined by amplitude, frequency content (where applicable), duty cycle, duration, and pulse spacing. Non-limiting pattern families include: 1. single pulse acknowledgment; 2. double pulse warning; 3. triple pulse adjustment request; 4. ramped pulses indicating progress toward acceptable coupling/quality; 5. long pulse + short pulses indicating fault states; and 6. timed sequences for guided protocols (e.g., baseline → actuation → recovery). Patterns may be configurable, may be disabled, and may be designed to be discreet and minimally audible.

F. Control Logic Driven by Sensor Fusion (Non-Limiting)

In various embodiments, haptic triggers and patterns are driven by one or more derived indices including: 1. Contact Integrity Index derived from pressure/contact sensing and strain stability; 2. PPG Coupling Quality Index derived from optical amplitude, SNR, and pulse plausibility; 3. Motion Stability Index derived from IMU jerk, orientation stability, and movement classification; 4. Closure/Seal Confirmation derived from hall effect/magnet detection and continuity checks; and 5. Composite Validity Score combining indices to determine when windows are valid, uncertain, or invalid. In certain embodiments, prompts are issued only when the system determines the user can correct the condition (e.g., during setup, when coupling repeatedly fails, or when motion stabilizes).

G. Optional Arousal Activation Protocols and Response Tagging (Non-Limiting)

In various embodiments, the device supports one or more user-enabled actuation protocols, including: 1. timed pulse protocols (e.g., pulsed bursts, stepped patterns, ramp-up/ramp-down); 2. context-aware protocols that adjust actuation intensity or cadence based on sensor-derived stability (e.g., only actuate when IMU indicates low motion and coupling quality is adequate); and 3. closed-loop protocols that adjust actuation within safety limits based on measured response features (e.g., change-from-baseline in PPG-derived features, strain/pressure trends, temperature trends), while preserving privacy and avoiding explicit anatomical imaging. In certain embodiments, the system computes response features relative to a baseline window (pre-actuation), an actuation window, and a recovery window, and reports the outputs as indices and confidence indicators.

H. Artifact Avoidance, Measurement Gating, and Labeling (Non-Limiting)

In various embodiments, the device mitigates measurement contamination from haptic actuation by: 1. gating: excluding samples during vibration pulses and/or immediate settling intervals; 2. flagging: labeling actuation windows so downstream algorithms avoid false inference; 3. timing control: actuating between sampling bursts or during less sensitive intervals; and/or 4. baseline re-stabilization: re-establishing strain/pressure baselines after actuation using post-actuation settling windows. In certain embodiments, any derived metrics computed across windows are confidence-weighted based on whether actuation occurred and on quality indices.

I. Safety, Comfort, Consent, and Power Management (Non-Limiting)

In various embodiments, vibration output is bounded by device firmware and hardware constraints including: 1. maximum amplitude and duty cycle limits; 2. maximum continuous duration and maximum cumulative runtime per interval; 3. user-configurable enable/disable controls; 4. automatic reduction or disablement in sleep-tagged windows (unless explicitly enabled); 5. thermal and battery-aware limitations; and 6. activation-mode constraints, including explicit user initiation, abort/stop controls, and automatic disablement upon detection of integrity faults or comfort-related thresholds.

J. Variations

Any embodiment described herein may omit haptics, substitute actuator types, modify patterns, alter placement, and/or adjust gating logic while maintaining the functional intent of providing discreet guidance, coupling confirmation, event marking, quality-aware feedback, and optional arousal activation protocols for an all-day/night physiology wearable.